

MAF-ID: Multi-Agent Framework for Interactive Dubbing through Deep Video Understanding

Zhanbin Hu
Beijing Information Science and
Technology University
Beijing, China
zhanbinhu@bistu.edu.cn

Xiaodong He
Wenzhou Institute of Industry and
Science
Wenzhou, China
dongms@zju.edu.cn

Renzhou Pan
Zhejiang Sci-Tech University
Hangzhou, China
panrz@mails.zstu.edu.cn

Xianzhou Zeng
Zhejiang University
Hangzhou, China
xzhouzeng@zju.edu.cn

Chenming Fan
Georgia Institute of Technology
Atlanta, United States
cfan70@gatech.edu

Qiang Zhu
Zhejiang University
Hangzhou, China
zhuq@zju.edu.cn

ABSTRACT

In the domain of video generation, Text-to-video[2] suffers from a notable application gap due to lack of audio that harmonizes with the visual content. Current solutions typically dubbing based solely on the original text used for generate video, which causes a mismatch between the video content and audio details, primarily stems from the lack of understanding of the video’s visual modality. Leveraging advancements in multimodal large language model[12] and LLM-based Agent[11], we propose MAF-ID, a multi-agent interactive framework for video dubbing based on deep video understanding. MAF-ID achieves agent collaboration through the autonomous interaction of three agents, to capture a deep understanding of the video visual content from macro to micro, progressively generate sound effects, voice-overs, and background music that is adaptive to the video. By deeply aligning text, video, and audio modalities, our method significantly enhances the fine-grained coordination between video and audio, making it widely available for AI-generated videos, VLOGs, and other video production scenarios requiring dubbing.

CCS CONCEPTS

• Computing methodologies → Scene understanding; • Human-centered computing → Auditory feedback.

KEYWORDS

Video Understanding; Multi-Agent; Audio Generation

ACM Reference Format:

Zhanbin Hu, Xiaodong He, Renzhou Pan, Xianzhou Zeng, Chenming Fan, and Qiang Zhu. 2024. MAF-ID: Multi-Agent Framework for Interactive Dubbing through Deep Video Understanding. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM ’24)*, October 28–November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3664647.3684992>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM ’24, October 28–November 1, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3684992>

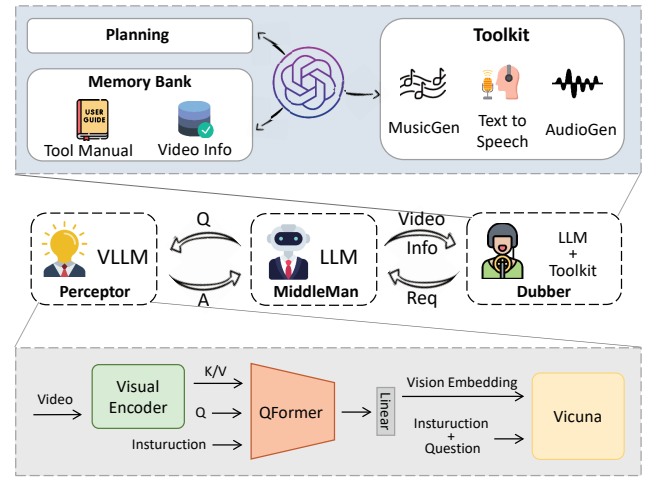


Figure 1: The design of MAF-ID’s Multi-Agent interaction architecture. The video perception agent Perceptor, coordinator agent MiddleMan, and dubbing agent Dubber interact independently, with video understanding guiding the dubbing for the video.

1 INTRODUCTION

The recent release of Sora[1] has sparked significant interest in the field of Text-to-Video[2]. Most research, such as Sora[1] and Open-Sora[13], based on DiT[5], focuses on optimizing text-to-video generation control and enhancing visual effects. However, sound, as a crucial medium for conveying information in videos, is often overlooked in these studies, limiting the completeness and expressiveness of the video content. Moreover, the diverse dubbing styles on the internet make it challenging to construct a training set that aligns video and audio effectively.

Recent advancements in AIGC-based Text-to-Music products like Suno[7] and Udio[8] have utilized internet music as training data to develop music generation models conditioned on text. Current solutions for dubbing Text-to-Video[2] videos mainly focused on using the text conditions for video generation to guide music generation. However, the vast space of video and music corresponding to the same text makes it difficult to achieve precise alignment.

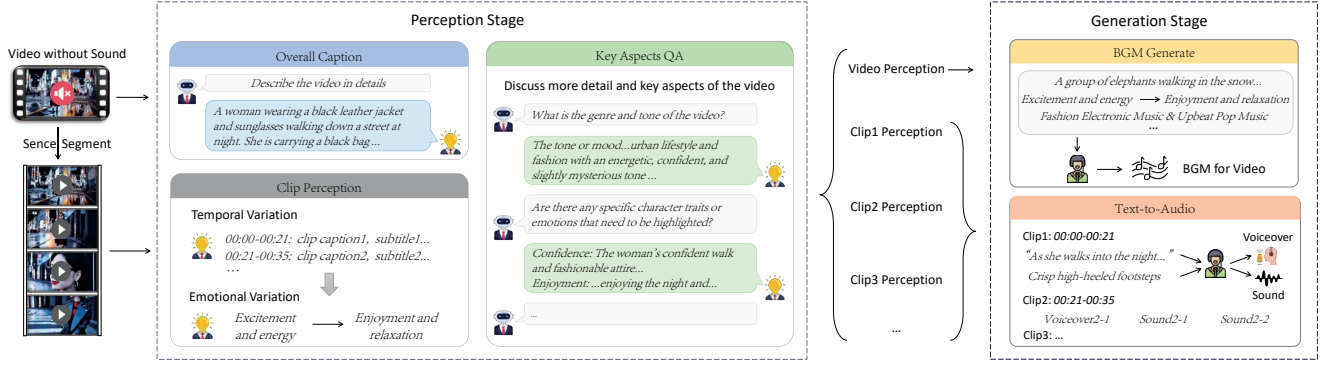


Figure 2: The overview of MAF-ID case mainly comprises two stages: the perception stage and the generation stage. In the perception stage, Perceptor interacts with Middleman to perceive the video content, while in the generation stage, Middleman interacts with Dubber to generate BGM, sound effects, and voice-overs.

Additionally, merely adding background music to AI-generated videos can make them monotonous and lack authenticity, necessitating the inclusion of sound effects or voice-overs suitable for the video scenes.

This paper introduces a multi-agent interactive video dubbing method, MAF-ID, guided by deep video understanding[6]. Through the autonomous interaction of three LLM-Based Agents[11]: the Video Perception Agent (Perceptor), the Coordinator Agent (MiddleMan), and the Dubbing Agent (Dubber), we achieve a deep understanding of video content from micro to macro levels. This understanding guides the generation of overall background music and segment-specific sound effects and voice-overs. MAF-ID has achieved the fusion of text, video, and audio modalities by multi-agent collaboration, ensures fine-grained alignment between video and generated audio, enhancing video expressiveness and applicability in various dubbing-required scenarios, including AI-generated videos, video editing, and Vlog creation.

2 MAF-ID

The Multi-Agent interaction architecture of MAF-ID, as shown in Figure 1, includes three agents: Perceptor, Middleman, and Dubber. Initially, Perceptor performs preliminary perception of the video. Middleman then queries Perceptor about specific video details and dubbing-related questions based on the perceived content. Dubber assesses whether the perceived information is sufficient for dubbing; if not, it requests further information from Middleman, who in turn queries Perceptor. Eventually, Dubber generates the background music based on a global understanding of the video and creates segment-specific sound effects and narrations based on detailed clip understanding.

Perceptor is responsible for video perception, adopting a typical multimodal large language model[12] architecture, comprising Visual Encoder, Qformer[4], and LLM, to query video content based on text prompts. We predefine some video content query questions, while others are dynamically proposed by MiddleMan.

Middleman coordinates between Perceptor and Dubber, constructed with an OpenAI LLM integrated with Chain-of-Thought (CoT)[10]. It is instructed to pose video understanding questions

to Perceptor, analyze the video information perceived by Perceptor, and plan more granular video-related questions for further discussion with Perceptor. Ultimately, Middleman summarizes the gathered information and conveys it to Dubber.

Dubber, an OpenAI Agent with task planning and tool-using capabilities[9], equipped with a Toolkit for audio generation, including MusicGen, AudioGen, and Text-to-Speech. Emphasizing video understanding-guided dubbing, we adopt Facebook’s open-source AudioCraft[3] rather than designing a new Text-to-Audio model. Dubber’s audio generation tools support hot-swapping, allowing users to replace them with other models like Suno[7] or Udio[8] freely. A Memory Bank[14] is added to Dubber to store tool usage instructions and cache received video information, enhancing the tool-using capabilities of Dubber.

3 USER EXPERIENCE

Figure 2 is an overview of a MAF-ID case, including the perception stage and the generation stage. The silent video input by the user is divided into several video clips by a scene-level video segmentation algorithm. During the perception stage, Perceptor interacts with Middleman to perform global and clip-level video perception, including the overall caption of the original video, clip captions, clip emotion variation, subtitles and some other video information perception, and further perceive more fine-grained key aspects and dubbing-related info of video based on foresaid info. For generation stage, Dubber will evaluate whether the perceived information is enough to dubbing, if enough, it will generate background music, sound effects, and voice-overs, otherwise, it feeds back to MiddleMan and return to the perception stage for iterative perception.

For the purpose of user-friendliness in the interaction, the reasoning behind the discussion of agents and perception of each video clips will present to user, allowing for modifications to ensure more reliable and stable control over audio generation.

Our method can be widely applied to AI-generated videos, VLOGs and other scenarios that require background music, sound effects, and voice-overs for videos. For the future, we also consider encapsulating it as plugins integrated into video editing software for more convenient use.

REFERENCES

- [1] Tim Brooks, Bill Peebles, Connor Holmes, Will DePue, Yufei Guo, Li Jing, David Schnurr, Joe Taylor, Troy Luhman, Eric Luhman, Clarence Ng, Ricky Wang, and Aditya Ramesh. 2024. Video generation models as world simulators. (2024). <https://openai.com/research/video-generation-models-as-world-simulators>
- [2] Joseph Cho, Fachrina Dewi Puspitasari, Sheng Zheng, Jingyao Zheng, Lik-Hang Lee, Tae-Ho Kim, Choong Seon Hong, and Chaoning Zhang. 2024. Sora as an agi world model? a complete survey on text-to-video generation. *arXiv preprint arXiv:2403.05131* (2024).
- [3] Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, Yossi Adi, and Alexandre Défossez. 2024. Simple and controllable music generation. *Advances in Neural Information Processing Systems* 36 (2024).
- [4] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*. PMLR, 19730–19742.
- [5] William Peebles and Saining Xie. 2023. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4195–4205.
- [6] Ghazala Rafiq, Muhammad Rafiq, and Gyu Sang Choi. 2023. Video description: A comprehensive survey of deep learning approaches. *Artificial Intelligence Review* 56, 11 (2023), 13293–13372.
- [7] Suno Team. 2024. Suno AI. <https://suno.com>. Accessed: 2024-06-01.
- [8] Udio Team. 2024. Udio AI Music Generator. <https://www.udio.com>. Accessed: 2024-06-01.
- [9] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, et al. 2024. A survey on large language model based autonomous agents. *Frontiers of Computer Science* 18, 6 (2024), 1–26.
- [10] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- [11] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. 2023. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864* (2023).
- [12] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. 2023. A survey on multimodal large language models. *arXiv preprint arXiv:2306.13549* (2023).
- [13] Zangwei Zheng, Xiangyu Peng, and Yang You. 2024. *Open-Sora: Democratizing Efficient Video Production for All*. <https://github.com/hpcaitech/Open-Sora>
- [14] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. Memorybank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 19724–19731.